NoisePrivacyMap<L2Distance<RBig>, ZeroConcentratedDivergence> for ZExpFamily<2>

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `NoisePrivacyMap` for `ZExpFamily<2>` in `mod.rs` at commit f5bb719 (outdated[1]).

1 Hoare Triple

Precondition

Compiler-Verified

`NoisePrivacyMap` is parameterized as follows:

- `MI`, the input metric, is of type `L2Distance<RBig>`
- `MO`, the output measure, is of type `ZeroConcentratedDivergence`

User-Verified

None

Pseudocode

```
# analogous to impl NoisePrivacyMap<L2Distance<RBig>, ZeroConcentratedDivergence> for ZExpFamily<1> in Rust
class ZExpFamily2:
    def noise_privacy_map(
        self, _input_metric: L2Distance[RBig], _output_measure: ZeroConcentratedDivergence
    ) -> PrivacyMap[L2Distance[RBig], ZeroConcentratedDivergence]:
        scale = self.scale
        if scale < RBig.ZERO:  #
            raise "scale must be non-negative"

        def privacy_map(d_in: RBig):
            if d_in < RBig.ZERO:  #
                raise "sensitivity must be non-negative"

            if d_in.is_zero():  #
                return 0.0

            if scale.is_zero():  #
                return float("inf")

```

[1] See new changes with `git diff f5bb719..785ed175 rust/src/measurements/noise/distribution/gaussian/mod.rs`

```
20            return f64.inf_cast((d_in / scale).pow(2) / rbig(2))  #
21
22        return PrivacyMap.new_fallible(privacy_map)
```

Postcondition

Theorem 1.1. Given a distribution `self`, returns `Err(e)` if `self` is not a valid distribution. Otherwise the output is `Ok(privacy_map)` where `privacy_map` observes the following:

Define `function(x) = x + Z` where `Z` is a vector of iid samples from `self`.

For every pair of elements x, x' in `VectorDomain<AtomDomain<IBig>>`, and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_measure`, if x, x' are `d_in`-close under `input_metric`, `privacy_map(d_in)` does not raise an exception, and `privacy_map(d_in)` $\leq$ `d_out`, then `function(x)`, `function(x')` are `d_out`-close under `output_measure`.

Proof. Line 7 rejects `self` if `self` does not represent a valid distribution, satisfying the error conditions of the postcondition.

We now construct the privacy map. First consider the extreme values of the scale and sensitivity parameters. The sensitivity `d_in`, a bound on distances, must not be negative, as checked on line 11. In the case where sensitivity is zero (line 14), the privacy loss is zero, regardless the choice of scale parameter (even zero). This is because the privacy loss when adjacent datasets are always identical is zero. Otherwise, in the case where the scale is zero, the privacy loss is infinite. To avoid a rational division overflow, line 17 returns infinity.

By line 20, both the sensitivity and scale are positive rationals. Write σ for `self.scale`.

If the metric is non-modular, Theorem 14 from [CKS20] implies that for all $\alpha > 1$,

$$D_\alpha(M(x), M(x')) \leq \alpha \cdot \frac{(\texttt{d_in}/\sigma)^2}{2}.$$

Suppose instead that the metric is modular. Let

$$G = \mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_d}$$

be the quotient group induced by the coordinatewise moduli, and let $\pi : \mathbb{Z}^d \to G$ be coordinatewise reduction modulo those moduli. The wrapped mechanism is

$$M_m(x) = \pi(x + Y),$$

where the coordinates of Y are independent discrete-Gaussian random variables with common scale σ.

Consider neighboring inputs $x \sim x'$ and let $r = x' - x \in G$. By the definition of modular L_2 distance, there exists a lift $t \in \mathbb{Z}^d$ such that $\pi(t) = r$ and

$$\sum_{i=1}^{d} \frac{t_i^2}{\sigma^2} \leq \left(\frac{\texttt{d_in}}{\sigma}\right)^2.$$

Choose any lift $a \in \mathbb{Z}^d$ of x. Then $a + t$ is a lift of x'. Let P be the law of $a + Y$ on $\mathbb{Z}^d$, and Q the law of $a + t + Y$. Their pushforwards under π are exactly the laws of $M_m(x)$ and $M_m(x')$. By data processing for Rényi divergence under the deterministic map π,

$$D_\alpha(M_m(x), M_m(x')) \leq D_\alpha(P, Q).$$

Since adding the same constant does not change Rényi divergence,

$$D_\alpha(P, Q) = D_\alpha(Y, Y + t).$$

Proposition 5 of [CKS20] gives the scalar discrete-Gaussian Rényi bound, and Theorem 14 gives the multivariate independent-coordinate version. Therefore, for all $\alpha > 1$,

$$D_\alpha(Y, Y + t) \leq \alpha \sum_{i=1}^{d} \frac{t_i^2}{2\sigma^2} \leq \alpha \cdot \frac{(\texttt{d_in}/\sigma)^2}{2}.$$

Combining the displays yields the same bound for the wrapped mechanism.

Line 20 implements this upper bound exactly as `d_in^2 / (2 * scale^2)` using exact rational arithmetic and a conservative cast to float. □

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.